



RA8876/RA8877

AP Note_ Source Code 001_How to initial RA8876,RA8877.

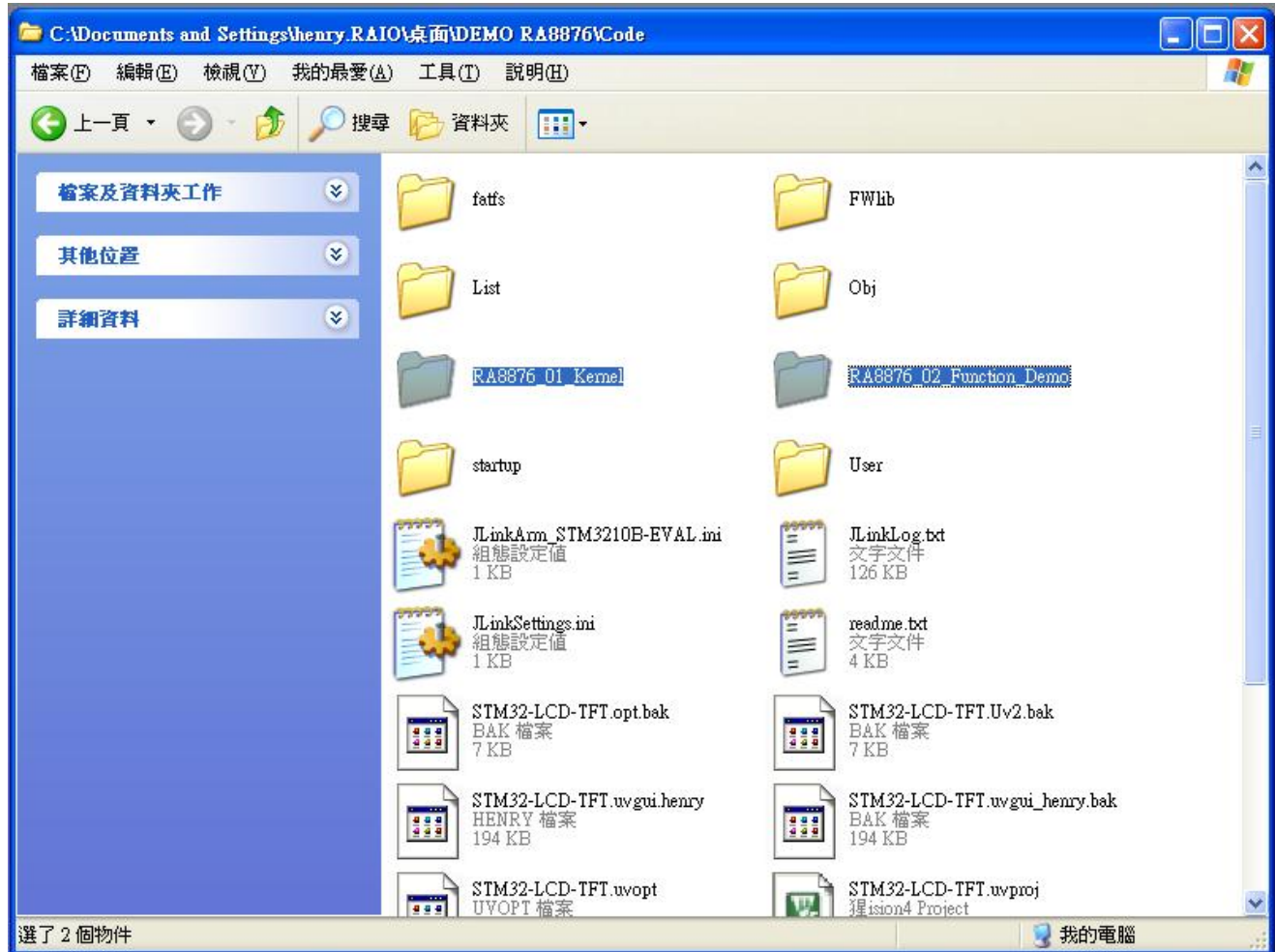
Revise History		
Version	Date	Description
0.1		

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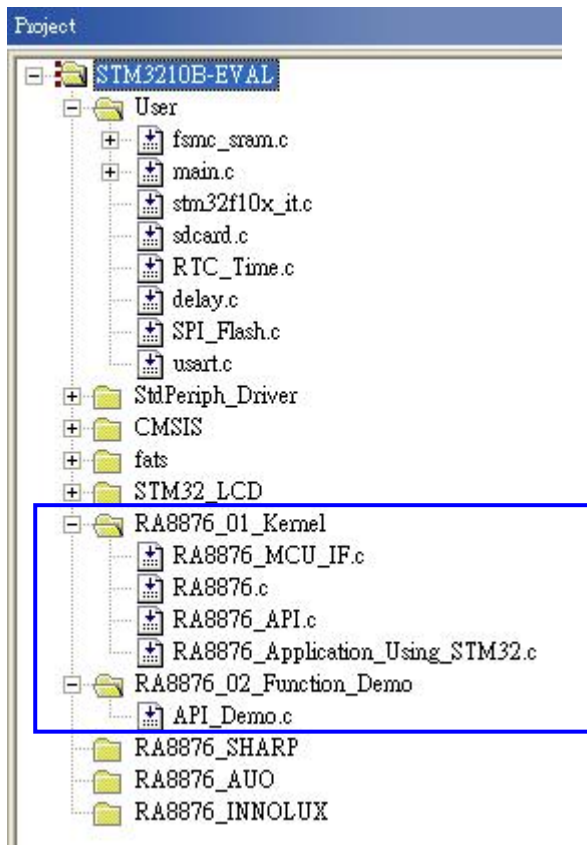
1. Flow

1.1 Please add the folder [RA8876_01_Kernel] and [RA8876_02_Function_Demo] in your project.



[RA8876_01_Kernel] :		
1	UserDef.h	
2	RA8876.c	the subroutine of all register.
3	RA8876_API.c	Application Programming Interface
4	RA8876_Application_Using_STM32.c	ex: STM32 read SD Card then write to RA8876(SDRAM).
5	RA8876_MCU_IF.c	
[RA8876_02_Function_Demo] :		
1	API_Demo.c	using <RA8876_API> to demo some functions.

1.2 Add files to Program editor.



1.3 Main

```
int main(void)
{
    delay_ms(10);
    RA8876_Initial();

    /*RA8876_02_Function_Demo*/

    API_Demo();
}
```

The starting speed of RA8876 is 107MHz (if OSC=10MHz), please assess the transmission speed of MCU Interface, ensure that RA8876 can receive instruction correctly .

2. UserDef.h

```
//-----
// Hardware Setting
//-----
/*==== [HW_(1)] MCU Interface =====*/
#define MEGAWIN_82G516A
#define STM_32
```

```
/*==== [HW_(2)] MCU I/F =====*/
#define Parallel_8080
// #define Parallel_6800
// #define Serial_3
// #define Serial_4
// #define Serial_I2C
```

Please modify the file
[RA8876_MCU_IF.c] and
[RA8876_MCU_IF.h] to match your use.

```
/*==== [HW_(3)] SDRAM =====*/
// #define IS42SM16160D
#define IS42S16320B
// #define IS42S16400F
// #define M12L32162A
// #define M12L2561616A
// #define M12L64164A
// #define W9825G6JH
// #define W9812G6JH
// #define MT48LC4M16A
// #define K4S641632N
// #define K4S281632K
```

(1) Please select a SDRAM or create a new one.
(2) Please check or modify the subroutine
[void RA8876_SDRAM_initail(void)] in
[RA8876.c].

```

/*==== [HW_(4)] Panel  =====*/
    #define EJ080NA_04B          ///INNOLUX_1024x768_EJ080NA-04B
//  #define LQ150X1LGN2C        ///SHARP_LQ150X1LGN2C_LVDS1 & LVDS2
//  #define N070ICG_LD1          ///INNOLUX_1280x800_N070ICG-LD1
//  #define B116XW03_V0
//  #define LQ121S1LG81
//  #define LQ035NC111
//  #define AT080TN52
//  #define ET101000DM6          ///EDT_1024x600_ET101000DM6
//  #define G190SVT01            ///AUO_1680x342_G190SVT01
//  #define ZJ070NA_01B          ///INNOLUX_1024x600_ZJ070NA_01B
//  #define LQ190E1LW52          ///SHARP_1280x1024_LQ190E1LW52
//  #define HJ070IA_02F          ///INNOLUX_1280x800_HJ070IA_02F
    
```

(1) Please select a LCD or create a new one.
 (2) Please check or modify the subroutine
 [void Set_LCD_Panel (void)] in [RA8876.c].

```

/*==== [HW_(5)] Serial Flash Memory */
    #define MX25L25635E

/*==== [HW_(6)] GT_Font ROM*/
//  #define GT21L16T1W
//  #define GT30L16U2W
    #define GT30L24T3Y
//  #define GT30L24M1Z
//  #define GT30L32S4W
//  #define GT20L24F6Y
//  #define GT21L24S1W
    
```

(1) Please select the SPI serial flash memory or
 create a new one.
 (2) Please check the subroutine [void
 Set_Serial_Flash_IF (void)] in [RA8876.c].

```
/*==== [SW_(1)] PLL =====*/
```

```
#define OSC_FREQ      10      // OSC clock frequency, unit: MHz.
#define DRAM_FREQ     120     // SDRAM clock frequency, unit: MHz.
#define CORE_FREQ     120     // Core (system) clock frequency, unit: MHz.
#define SCAN_FREQ     58      // Panel Scan clock frequency, unit: MHz.
```

Please depending on the application needs, set the PLL frequency.

The limits of PLL are as follows :

(1)

RA8876 / RA8877 datasheet :			
SCAN_FREQ	CORE_FREQ		DRAM_FREQ
max 100 MHz	with internal font	max 120 MHz	max 166 MHz
	without internal font	max 133 MHz	

(2)

conditions :				
16bit SDRAM				
Scan output 8bpp	$SCAN_FREQ \leq 100MHz$	<	$CORE_FREQ$	$\leq DRAM_FREQ$
Scan output 16bpp	$SCAN_FREQ \leq (CORE_FREQ / 1.5)$	<	$CORE_FREQ$	$\leq DRAM_FREQ$
Scan output 24bpp	$SCAN_FREQ \leq (CORE_FREQ / 2)$	<	$CORE_FREQ$	$\leq DRAM_FREQ$
32bit SDRAM				
Scan output 8bpp	$SCAN_FREQ \leq 100MHz$	<	$CORE_FREQ$	$\leq DRAM_FREQ$
Scan output 16bpp	$SCAN_FREQ \leq 100MHz$	<	$CORE_FREQ$	$\leq DRAM_FREQ$
Scan output 24bpp	$SCAN_FREQ \leq 100MHz$	<	$CORE_FREQ$	$\leq DRAM_FREQ$

LCD datasheet :

Parameter		Symbol	Min.	Typ.	Max.	Unit
Clock signal	Frequency	1/Tc	50.0	65.0	80.0	MHz
ENAB signal	Horizontal period	TH	1056	1344	1720	clock
			16.0	20.7	23.4	μs
	Horizontal period (High)	THd	1024	1024	1024	clock
	Vertical period	TV	773	806	990	line
	Vertical period (High)	TVd	768	768	768	line

```
/*==== [SW_(2)] select MCU and color depth =====*/
```

```
#define MCU_8bit_ColorDepth_8bpp
```

```
// #define MCU_8bit_ColorDepth_16bpp
```

```
// #define MCU_8bit_ColorDepth_24bpp
```

```
// #define MCU_16bit_ColorDepth_16bpp
```

```
// #define MCU_16bit_ColorDepth_24bpp_Mode_1
```

```
// #define MCU_16bit_ColorDepth_24bpp_Mode_2
```

Depending on application and content to define the color depth.

MCU I/F

The data format of writing SDRAM.

8-bit MPU, 8bpp mode (RGB 3:3:2)

Order	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
1	R ₀ ⁷	R ₀ ⁶	R ₀ ⁵	G ₀ ⁷	G ₀ ⁶	G ₀ ⁵	B ₀ ⁷	B ₀ ⁶
2	R ₁ ⁷	R ₁ ⁶	R ₁ ⁵	G ₁ ⁷	G ₁ ⁶	G ₁ ⁵	B ₁ ⁷	B ₁ ⁶
3	R ₂ ⁷	R ₂ ⁶	R ₂ ⁵	G ₂ ⁷	G ₂ ⁶	G ₂ ⁵	B ₂ ⁷	B ₂ ⁶
4	R ₃ ⁷	R ₃ ⁶	R ₃ ⁵	G ₃ ⁷	G ₃ ⁶	G ₃ ⁵	B ₃ ⁷	B ₃ ⁶
5	R ₄ ⁷	R ₄ ⁶	R ₄ ⁵	G ₄ ⁷	G ₄ ⁶	G ₄ ⁵	B ₄ ⁷	B ₄ ⁶
6	R ₅ ⁷	R ₅ ⁶	R ₅ ⁵	G ₅ ⁷	G ₅ ⁶	G ₅ ⁵	B ₅ ⁷	B ₅ ⁶

8-bit MPU, 16bpp mode (RGB 5:6:5)

Order	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
1	G ₀ ⁴	G ₀ ³	G ₀ ²	B ₀ ⁷	B ₀ ⁶	B ₀ ⁵	B ₀ ⁴	B ₀ ³
2	R ₀ ⁷	R ₀ ⁶	R ₀ ⁵	R ₀ ⁴	R ₀ ³	G ₀ ⁷	G ₀ ⁶	G ₀ ⁵
3	G ₁ ⁴	G ₁ ³	G ₁ ²	B ₁ ⁷	B ₁ ⁶	B ₁ ⁵	B ₁ ⁴	B ₁ ³
4	R ₁ ⁷	R ₁ ⁶	R ₁ ⁵	R ₁ ⁴	R ₁ ³	G ₁ ⁷	G ₁ ⁶	G ₁ ⁵
5	G ₂ ⁴	G ₂ ³	G ₂ ²	B ₂ ⁷	B ₂ ⁶	B ₂ ⁵	B ₂ ⁴	B ₂ ³
6	R ₂ ⁷	R ₂ ⁶	R ₂ ⁵	R ₂ ⁴	R ₂ ³	G ₂ ⁷	G ₂ ⁶	G ₂ ⁵

8-bit MPU, 24bpp mode (RGB 8:8:8)

Order	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
1	B ₀ ⁷	B ₀ ⁶	B ₀ ⁵	B ₀ ⁴	B ₀ ³	B ₀ ²	B ₀ ¹	B ₀ ⁰
2	G ₀ ⁷	G ₀ ⁶	G ₀ ⁵	G ₀ ⁴	G ₀ ³	G ₀ ²	G ₀ ¹	G ₀ ⁰
3	R ₀ ⁷	R ₀ ⁶	R ₀ ⁵	R ₀ ⁴	R ₀ ³	R ₀ ²	R ₀ ¹	R ₀ ⁰
4	B ₁ ⁷	B ₁ ⁶	B ₁ ⁵	B ₁ ⁴	B ₁ ³	B ₁ ²	B ₁ ¹	B ₁ ⁰
5	G ₁ ⁷	G ₁ ⁶	G ₁ ⁵	G ₁ ⁴	G ₁ ³	G ₁ ²	G ₁ ¹	G ₁ ⁰
6	R ₁ ⁷	R ₁ ⁶	R ₁ ⁵	R ₁ ⁴	R ₁ ³	R ₁ ²	R ₁ ¹	R ₁ ⁰

16-bit MPU, 8bpp mode 1 (RGB 3:3:2)

Order	Bit15	Bit14	Bit13	Bit12	Bit11	Bit10	Bit9	Bit8	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
1	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	R ₀ ⁷	R ₀ ⁶	R ₀ ⁵	G ₀ ⁷	G ₀ ⁶	G ₀ ⁵	B ₀ ⁷	B ₀ ⁶
2	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	R ₁ ⁷	R ₁ ⁶	R ₁ ⁵	G ₁ ⁷	G ₁ ⁶	G ₁ ⁵	B ₁ ⁷	B ₁ ⁶
3	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	R ₂ ⁷	R ₂ ⁶	R ₂ ⁵	G ₂ ⁷	G ₂ ⁶	G ₂ ⁵	B ₂ ⁷	B ₂ ⁶
4	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	R ₃ ⁷	R ₃ ⁶	R ₃ ⁵	G ₃ ⁷	G ₃ ⁶	G ₃ ⁵	B ₃ ⁷	B ₃ ⁶
5	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	R ₄ ⁷	R ₄ ⁶	R ₄ ⁵	G ₄ ⁷	G ₄ ⁶	G ₄ ⁵	B ₄ ⁷	B ₄ ⁶
6	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	R ₅ ⁷	R ₅ ⁶	R ₅ ⁵	G ₅ ⁷	G ₅ ⁶	G ₅ ⁵	B ₅ ⁷	B ₅ ⁶

16-bit MPU, 16bpp mode (RGB 5:6:5)

Order	Bit15	Bit14	Bit13	Bit12	Bit11	Bit10	Bit9	Bit8	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
1	R ₀ ⁷	R ₀ ⁶	R ₀ ⁵	R ₀ ⁴	R ₀ ³	G ₀ ⁷	G ₀ ⁶	G ₀ ⁵	G ₀ ⁴	G ₀ ³	G ₀ ²	B ₀ ⁷	B ₀ ⁶	B ₀ ⁵	B ₀ ⁴	B ₀ ³
2	R ₁ ⁷	R ₁ ⁶	R ₁ ⁵	R ₁ ⁴	R ₁ ³	G ₁ ⁷	G ₁ ⁶	G ₁ ⁵	G ₁ ⁴	G ₁ ³	G ₁ ²	B ₁ ⁷	B ₁ ⁶	B ₁ ⁵	B ₁ ⁴	B ₁ ³
3	R ₂ ⁷	R ₂ ⁶	R ₂ ⁵	R ₂ ⁴	R ₂ ³	G ₂ ⁷	G ₂ ⁶	G ₂ ⁵	G ₂ ⁴	G ₂ ³	G ₂ ²	B ₂ ⁷	B ₂ ⁶	B ₂ ⁵	B ₂ ⁴	B ₂ ³
4	R ₃ ⁷	R ₃ ⁶	R ₃ ⁵	R ₃ ⁴	R ₃ ³	G ₃ ⁷	G ₃ ⁶	G ₃ ⁵	G ₃ ⁴	G ₃ ³	G ₃ ²	B ₃ ⁷	B ₃ ⁶	B ₃ ⁵	B ₃ ⁴	B ₃ ³
5	R ₄ ⁷	R ₄ ⁶	R ₄ ⁵	R ₄ ⁴	R ₄ ³	G ₄ ⁷	G ₄ ⁶	G ₄ ⁵	G ₄ ⁴	G ₄ ³	G ₄ ²	B ₄ ⁷	B ₄ ⁶	B ₄ ⁵	B ₄ ⁴	B ₄ ³
6	R ₅ ⁷	R ₅ ⁶	R ₅ ⁵	R ₅ ⁴	R ₅ ³	G ₅ ⁷	G ₅ ⁶	G ₅ ⁵	G ₅ ⁴	G ₅ ³	G ₅ ²	B ₅ ⁷	B ₅ ⁶	B ₅ ⁵	B ₅ ⁴	B ₅ ³

16-bit MPU, 24bpp mode 1 (RGB 8:8:8)

Order	Bit15	Bit14	Bit13	Bit12	Bit11	Bit10	Bit9	Bit8	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
1	G ₀ ⁷	G ₀ ⁶	G ₀ ⁵	G ₀ ⁴	G ₀ ³	G ₀ ²	G ₀ ¹	G ₀ ⁰	B ₀ ⁷	B ₀ ⁶	B ₀ ⁵	B ₀ ⁴	B ₀ ³	B ₀ ²	B ₀ ¹	B ₀ ⁰
2	B ₁ ⁷	B ₁ ⁶	B ₁ ⁵	B ₁ ⁴	B ₁ ³	B ₁ ²	B ₁ ¹	B ₁ ⁰	R ₀ ⁷	R ₀ ⁶	R ₀ ⁵	R ₀ ⁴	R ₀ ³	R ₀ ²	R ₀ ¹	R ₀ ⁰
3	R ₁ ⁷	R ₁ ⁶	R ₁ ⁵	R ₁ ⁴	R ₁ ³	R ₁ ²	R ₁ ¹	R ₁ ⁰	G ₁ ⁷	G ₁ ⁶	G ₁ ⁵	G ₁ ⁴	G ₁ ³	G ₁ ²	G ₁ ¹	G ₁ ⁰
4	G ₂ ⁷	G ₂ ⁶	G ₂ ⁵	G ₂ ⁴	G ₂ ³	G ₂ ²	G ₂ ¹	G ₂ ⁰	B ₂ ⁷	B ₂ ⁶	B ₂ ⁵	B ₂ ⁴	B ₂ ³	B ₂ ²	B ₂ ¹	B ₂ ⁰
5	B ₃ ⁷	B ₃ ⁶	B ₃ ⁵	B ₃ ⁴	B ₃ ³	B ₃ ²	B ₃ ¹	B ₃ ⁰	R ₂ ⁷	R ₂ ⁶	R ₂ ⁵	R ₂ ⁴	R ₂ ³	R ₂ ²	R ₂ ¹	R ₂ ⁰
6	R ₃ ⁷	R ₃ ⁶	R ₃ ⁵	R ₃ ⁴	R ₃ ³	R ₃ ²	R ₃ ¹	R ₃ ⁰	G ₃ ⁷	G ₃ ⁶	G ₃ ⁵	G ₃ ⁴	G ₃ ³	G ₃ ²	G ₃ ¹	G ₃ ⁰

16-bit MPU, 24bpp mode 2 (RGB 8:8:8)

Order	Bit15	Bit14	Bit13	Bit12	Bit11	Bit10	Bit9	Bit8	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
1	G ₀ ⁷	G ₀ ⁶	G ₀ ⁵	G ₀ ⁴	G ₀ ³	G ₀ ²	G ₀ ¹	G ₀ ⁰	B ₀ ⁷	B ₀ ⁶	B ₀ ⁵	B ₀ ⁴	B ₀ ³	B ₀ ²	B ₀ ¹	B ₀ ⁰
2	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	R ₀ ⁷	R ₀ ⁶	R ₀ ⁵	R ₀ ⁴	R ₀ ³	R ₀ ²	R ₀ ¹	R ₀ ⁰
3	G ₁ ⁷	G ₁ ⁶	G ₁ ⁵	G ₁ ⁴	G ₁ ³	G ₁ ²	G ₁ ¹	G ₁ ⁰	B ₁ ⁷	B ₁ ⁶	B ₁ ⁵	B ₁ ⁴	B ₁ ³	B ₁ ²	B ₁ ¹	B ₁ ⁰
4	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	R ₁ ⁷	R ₁ ⁶	R ₁ ⁵	R ₁ ⁴	R ₁ ³	R ₁ ²	R ₁ ¹	R ₁ ⁰
5	G ₂ ⁷	G ₂ ⁶	G ₂ ⁵	G ₂ ⁴	G ₂ ³	G ₂ ²	G ₂ ¹	G ₂ ⁰	B ₂ ⁷	B ₂ ⁶	B ₂ ⁵	B ₂ ⁴	B ₂ ³	B ₂ ²	B ₂ ¹	B ₂ ⁰
6	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	R ₂ ⁷	R ₂ ⁶	R ₂ ⁵	R ₂ ⁴	R ₂ ³	R ₂ ²	R ₂ ¹	R ₂ ⁰

3. RA8876.C

3.1 void RA8876_Initial(void)

```

void RA8876_Initial(void)
{
//-----
#ifdef Parallel_8080
    /* using STM32 FSMC Interface*/
#endif
#ifdef Parallel_6800
    Parallel_6800_ini();
#endif
#ifdef Serial_I2C
    I2C_Init();
#endif
#ifdef Serial_SPI4
    SPI4_Init();
#endif
#ifdef Serial_SPI3
    SPI3_Init();
#endif
//-----

    RA8876_HW_Reset();
//    RA8876_SW_Reset();
    Check_IC_ready();
//-----

    RA8876_PLL(); // Set RA8876/77 PLL and Start PLL
//-----
  
```

Initial MCU Interface

```

Ex:
void RA8876_PLL(void)
{
    //All PLL Divided by 4
    LCD_RegisterWrite(0x05,0x04);
    LCD_RegisterWrite(0x07,0x04);
    LCD_RegisterWrite(0x09,0x04);
    LCD_RegisterWrite(0x06,(SCAN_FREQ*4/10)-1);
    LCD_RegisterWrite(0x08,(DRAM_FREQ*4/10)-1);
    LCD_RegisterWrite(0x0A,(CORE_FREQ*4/10)-1);

    Enable_PLL();
    delay_ms(1);
}
  
```

```

[ UserDef.h ] :
/*==== [SW_(1)] PLL =====*/
#define OSC_FREQ 10
#define DRAM_FREQ 120
#define CORE_FREQ 120
#define SCAN_FREQ 58
  
```

```
//-----
RA8876_SDRAM_initial(); // set the SDRAM paramete
//-----
```

```
Ex:
void RA8876_SDRAM_initial(void)
{
#ifdef W9825G6JH
    CAS_Latency=3;
    Auto_Refresh=(64*DRAM_FREQ*1000)/(8192);

    LCD_RegisterWrite(0xe0,0x31);
    LCD_RegisterWrite(0xe1,CAS_Latency);
    LCD_RegisterWrite(0xe2,Auto_Refresh);
    LCD_RegisterWrite(0xe3,Auto_Refresh>>8);
    LCD_RegisterWrite(0xe4,0x01);
#endif
}
```

SDRAM :
3. AVAILABLE PART NUMBER

PART NUMBER	SPEED GRADE	SEL
W9825G6JH-5	200MHz/CL3	
W9825G6JH-6	166MHz/CL3 or 133MHz/CL2	
W9825G6JH-6I	166MHz/CL3	
W9825G6JH-6A	166MHz/CL3	
W9825G6JH-6L	166MHz/CL3	
W9825G6JH-75	133MHz/CL3	
W9825G6JH75L	133MHz/CL3	

SDRAM :

- 8K Refresh Cycles/64 mS
- Interface: LVTTTL
- Packaged in TSOP II 54-pin,

```
/**[01h]**/
```

```
TFT_24bit();//RA8876 only
```

```
// TFT_18bit();//RA8876 only
```

```
// TFT_16bit();//RA8876 only
```

```
// Without_TFT();//RA8876 only
```

```
//-----
```

Set RA8876 [LCD Panel Data Bus] output format.

LCD Panel Data Bus

TFT LCD data bus output for source driver. RA8876 supports 64K/256K/16.7M color depth by register setting; user can connect corresponding RGB bus for different setting.

Pin Name	Digital TFT Interface			
TFT output Setting	11b (GPIO)	10b (16-bits)	01b (18-bits)	00b (24-bits)
XPDAT[0]	GPIO-D0/ XKIN[1]			B0
XPDAT[1]	GPIO-D1/ XKIN[2]			B1
XPDAT[2]	GPIO-D6/ XKIN[4]		B0	B2
XPDAT[3]	GPIO-E0	B0	B1	B3
XPDAT[4]	GPIO-E1	B1	B2	B4
XPDAT[5]	GPIO-E2	B2	B3	B5
XPDAT[6]	GPIO-E3	B3	B4	B6
XPDAT[7]	GPIO-E4	B4	B5	B7
XPDAT[8]	GPIO-D2/ XKIN[3]			G0
XPDAT[9]	GPIO-D3/ XKOUT[3]			G1
XPDAT[10]	GPIO-E5	G0	G0	G2
XPDAT[11]	GPIO-E6	G1	G1	G3
XPDAT[12]	GPIO-E7	G2	G2	G4
XPDAT[13]	GPIO-F0	G3	G3	G5
XPDAT[14]	GPIO-F1	G4	G4	G6
XPDAT[15]	GPIO-F2	G5	G5	G7
XPDAT[16]	GPIO-D4/ XKOUT[1]			R0
XPDAT[17]	GPIO-D5/ XKOUT[2]			R1
XPDAT[18]	GPIO-D7/ XKOUT[4]		R0	R2
XPDAT[19]	GPIO-F3	R0	R1	R3
XPDAT[20]	GPIO-F4	R1	R2	R4
XPDAT[21]	GPIO-F5	R2	R3	R5
XPDAT[22]	GPIO-F6	R3	R4	R6
XPDAT[23]	GPIO-F7	R4	R5	R7

*unused pins can be programmed as GPIO-D/E/F(default) or XKIN/XOUT.

Default is 18bpp function mode, so XPDAT[17:16/8:9/1:0] are default at GPI mode.

```
/**[01h]**/
```

```
#if defined (MCU_8bit_ColorDepth_8bpp)
```

```
|| defined (MCU_8bit_ColorDepth_16bpp)
```

```
|| defined (MCU_8bit_ColorDepth_24bpp)
```

```
    Host_Bus_8bit();
```

```
#endif
```

(1) Select host data bus.

(2) If MCU I/F select SPI-3, SPI-4 or I2C, chip will only allow Host_Bus_8bit.

```
#if defined (MCU_16bit_ColorDepth_16bpp)
```

```
|| defined (MCU_16bit_ColorDepth_24bpp_Mode_1)
```

```
|| defined (MCU_16bit_ColorDepth_24bpp_Mode_2)
```

```
    Host_Bus_16bit();
```

```
#endif
```

/[02h]**/****#ifdef MCU_8bit_ColorDepth_8bpp****Data_Format_8b_8bpp();****#endif****#ifdef MCU_8bit_ColorDepth_16bpp****Data_Format_8b_16bpp();****#endif****#ifdef MCU_8bit_ColorDepth_24bpp****Data_Format_8b_24bpp();****#endif****#ifdef MCU_16bit_ColorDepth_16bpp****Data_Format_16b_16bpp();****#endif****#ifdef MCU_16bit_ColorDepth_24bpp_Mode_1****Data_Format_16b_24bpp_mode1();****#endif****#ifdef MCU_16bit_ColorDepth_24bpp_Mode_2****Data_Format_16b_24bpp_mode2();****#endif****MemWrite_Left_Right_Top_Down();//Memory Write: Left to Right, Top to Down.****/**[03h]**/****Graphic_Mode();****// Text_Mode();****Memory_Select_SDRAM();**

Select a data format for MCU
read/write SDRAM.
(ex: page 9 and page 10.)

```

/** Set color depth, define in [UserDef.h] */
#ifdef MCU_8bit_ColorDepth_8bpp
    Select_Main_Window_8bpp();           //[10h] Set main window color depth
    Memory_8bpp_Mode();                 //[5Eh] Set active memory color depth

    Select_PIP1_Window_8bpp();           //[11h] PIP 1 Window Color Depth
    Select_PIP2_Window_8bpp();           //[11h] PIP 2 Window Color Depth

    BTE_S0_Color_8bpp();                 //[92h] Source_0 Color Depth
    BTE_S1_Color_8bpp();                 //[92h] Source_1 Color Depth
    BTE_Destination_Color_8bpp();        //[92h] Destination Color Depth
#endif

#ifdef (MCU_8bit_ColorDepth_16bpp) || defined (MCU_16bit_ColorDepth_16bpp)
    Select_Main_Window_16bpp();          //[10h] Set main window color depth
    Memory_16bpp_Mode();                 //[5Eh]Set active memory color depth

    Select_PIP1_Window_16bpp();          //[11h] PIP 1 Window Color Depth
    Select_PIP2_Window_16bpp();          //[11h] PIP 2 Window Color Depth

    BTE_S0_Color_16bpp();                //[92h] Source_0 Color Depth
    BTE_S1_Color_16bpp();                //[92h] Source_1 Color Depth
    BTE_Destination_Color_16bpp();        //[92h] Destination Color Depth
#endif

#ifdef (MCU_8bit_ColorDepth_24bpp)
    || defined (MCU_16bit_ColorDepth_24bpp_Mode_1)
    || defined (MCU_16bit_ColorDepth_24bpp_Mode_2)
    Select_Main_Window_24bpp();          //[10h] Set main window color depth
    Memory_24bpp_Mode();                 //[5Eh]Set active memory color depth

    Select_PIP1_Window_24bpp();          //[11h] PIP 1 Window Color Depth
    Select_PIP2_Window_24bpp();          //[11h] PIP 2 Window Color Depth

    BTE_S0_Color_24bpp();                //[92h] Source_0 Color Depth
    BTE_S1_Color_24bpp();                //[92h] Source_1 Color Depth
    BTE_Destination_Color_24bpp();        //[92h] Destination Color Depth
#endif

```

```
/**[12h]~[1Fh]**//
```

```
Set_LCD_Panel(); // define in [UserDef.h]
```

Ex:

```
void Set_LCD_Panel(void)
{
```

```
    #ifdef HJ070IA_02F ///INNOLUX_1280x800_HJ070IA_02F
```

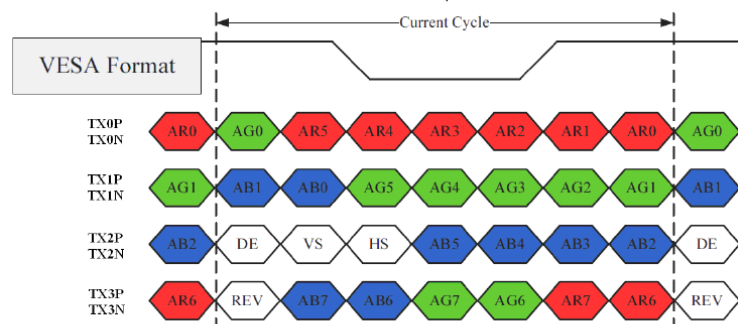
```
        /**[03h]**//
```

```
        LVDS_Format1(); //(VESA format) //only for RA8877
```

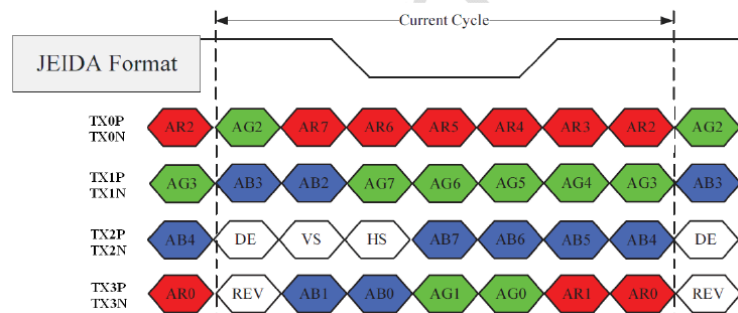
```
        // LVDS_Format2(); //(JEIDA format) //only for RA8877
```

according to LCD spec.

- Format 1 (VESA format): use with displays expecting the 2 MSB to be transmitted over the 4th data channel Y3. This is the dominate data format for LCD panels.



- Format 2 (JEIDA format): use with displays expecting the 2 LSB to be transmitted over the 4th data channel.



```

/**[10h]**/
    Select_LCD_Sync_Mode();    // Enable XVSYNCR, XHSYNCR, XDE.
    // Select_LCD_DE_Mode();    // XVSYNCR & XHSYNCR in idle state.

/**[12h]**/
    // PCLK_Rising();
    PCLK_Falling();
    //
    VSCAN_T_to_B();
    // VSCAN_B_to_T();
    //
    PDATA_Set_RGB();
    // PDATA_Set_RGB();
    // PDATA_Set_GRB();
    // PDATA_Set_GBR();
    // PDATA_Set_BRG();
    // PDATA_Set_BGR();
    // PDATA_Set_Gray();
    // PDATA_IDLE_STATE();

```

RA8876 LCD Panel Digital Interface																									
XPDAT	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
000b: RGB	R7	R6	R5	R4	R3	R2	R1	R0	G7	G6	G5	G4	G3	G2	G1	G0	B7	B6	B5	B4	B3	B2	B1	B0	
001b: RBG	R7	R6	R5	R4	R3	R2	R1	R0	B7	B6	B5	B4	B3	B2	B1	B0	G7	G6	G5	G4	G3	G2	G1	G0	
010b: GRB	G7	G6	G5	G4	G3	G2	G1	G0	R7	R6	R5	R4	R3	R2	R1	R0	B7	B6	B5	B4	B3	B2	B1	B0	
011b: GBR	G7	G6	G5	G4	G3	G2	G1	G0	B7	B6	B5	B4	B3	B2	B1	B0	R7	R6	R5	R4	R3	R2	R1	R0	
100b: BRG	B7	B6	B5	B4	B3	B2	B1	B0	R7	R6	R5	R4	R3	R2	R1	R0	G7	G6	G5	G4	G3	G2	G1	G0	
101b: BGR	B7	B6	B5	B4	B3	B2	B1	B0	G7	G6	G5	G4	G3	G2	G1	G0	R7	R6	R5	R4	R3	R2	R1	R0	

```

/**[13h]**/
    HSYNC_Low_Active();
// HSYNC_High_Active();

    VSYNC_Low_Active();
// VSYNC_High_Active();

// DE_Low_Active();
    DE_High_Active();

```

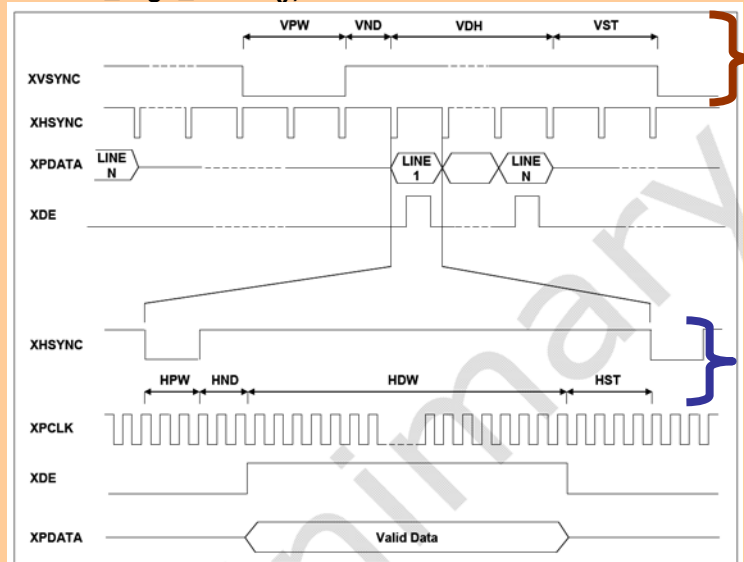


Figure 10-1 : Digital TFT Panel Timing

```

/**[14h][15h][1Ah][1Bh]**/
    LCD_HorizontalWidth_VeriticalHeight(1280,800);//1280x800
/**[16h][17h]**/
    LCD_Horizontal_Non_Display(30);//Blank=160 °
/**[18h]**/
    LCD_HSYNC_Start_Position(30);
/**[19h]**/
    LCD_HSYNC_Pulse_Width(100);
/**[1Ch][1Dh]**/
    LCD_Vertical_Non_Display(4);//Blank=23 °
/**[1Eh]**/
    LCD_VSYNC_Start_Position(4);
/**[1Fh]**/
    LCD_VSYNC_Pulse_Width(15);

```

#endif

}

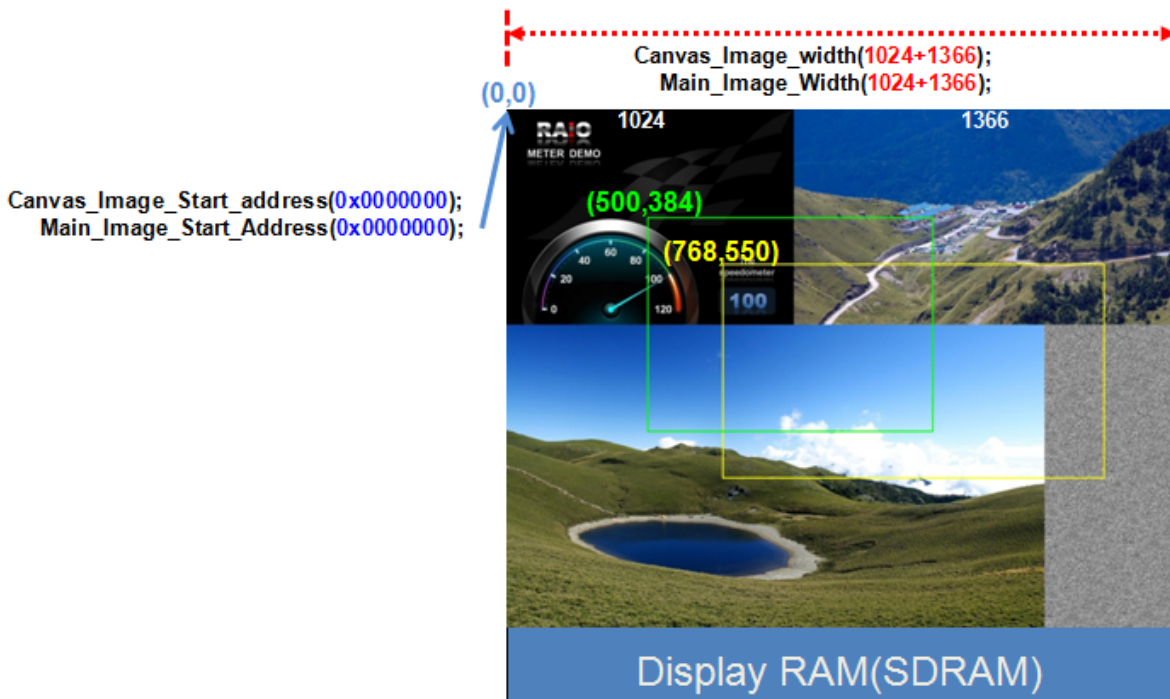
Ex:

Item	Symbol	Values			Unit	Remark
		Min.	Typ.	Max.		
Clock Frequency	fclk	(67.55)	(71.11)	(73)	MHz	Frame rate =60Hz
Horizontal display area	thd	1280				
HS period time	th	(1410)	(1440)	(1470)	DCLK	
HS Blanking	thb	130	160	190	DCLK	
Vertical display area	tvd	800				
VS period time	tv	(813)	(823)	(833)	H	
VS Blanking	thb	13	23	33	H	

```

/**[20h]~[25h]**//
    Main_Image_Start_Address(0);
    Main_Image_Width(1024*3);
/**[26h][27h][28h][29h]**//
    Main_Window_Start_XY(0,0);
/**[50h]~[55h]**//
    Canvas_Image_Start_address(0);
    Canvas_image_width(1024*3);
/**[56h][57h][58h][59h]**//
    Active_Window_XY(0,0);
    Active_Window_WH(1024,768);
  
```

Configure Main Image and
Canvas Image parameter.



```

( If LCD = 1024x768 )
Canvas_Image_Start_address(0x00000000);
Canvas_Image_width(1024+1366);
Main_Image_Start_Address(0x00000000);
Main_Image_Width(1024+1366);
Main_Window_Start_XY(500,384);
  
```



```

( If LCD = 1366x768 )
Canvas_Image_Start_address(0x00000000);
Canvas_Image_width(1024+1366);
Main_Image_Start_Address(0x00000000);
Main_Image_Width(1024+1366);
Main_Window_Start_XY(768,550);
  
```



```
/**[5E]**/
```

```
Memory_XY_Mode();
```

```
// Memory_Linear_Mode();
```

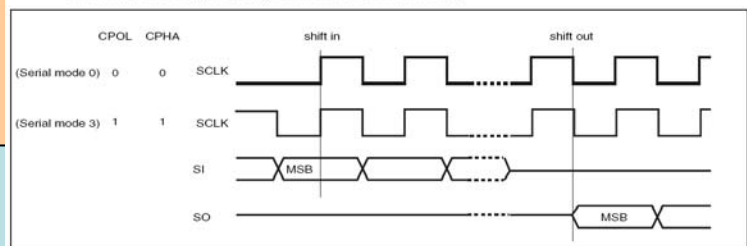
```
//-----
```

```
Set_Serial_Flash_IF();
```

```
//-----
```

```
Ex:
void Set_Serial_Flash_IF(void)
{   #ifdef MX25L25635E
    Enable_SFlash_SPI();
    #endif
    /**(*) Set RA8876/77 Master SPI address width.
    Select_SFI_24bit_Address();
    //Select_SFI_32bit_Address();
    /**(*)
    Select_standard_SPI_Mode0_or_Mode3();
    //Select_RA8875_SPI_Mode0_and_Mode3();
    /**(*)
    Select_SFI_Single_Mode_Dummy_0T_03h();
    // Select_SFI_Single_Mode_Dummy_8T_0Bh();
    // Select_SFI_Single_Mode_Dummy_16T_1Bh();
    // Select_SFI_Dual_Mode_Dummy_8T_3Bh();
    // Select_SFI_Dual_Mode_Dummy_4T_BBh();
    /**(*)
    Reset_CPOL();
    // Set_CPOL();
    Reset_CPHA();
    // Set_CPHA();
}
```

Serial Modes Supported (for Normal Serial mode)



```
/**[5Fh][60h][61h][62h]**/
```

```
Goto_Pixel_XY(0,0);
```

```
//-----
```

```
Display_ON();
```

```
Color_Bar_ON();
```

```
delay_seconds(1);
```

```
Color_Bar_OFF();
```

```
}
```

Thank you